FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 2

B.Sc. MATHEMATICS HONOURS

MAT5EJ303 (2): CONVEX OPTIMIZATION

Modules I–II: Stationary Points, Hessian Matrix, Taylor's Theorem, Classification of Stationary Points; Affine and Convex Sets, Convexity-Preserving Operations, Generalized Inequalities

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Define a stationary point of a real-valued function f of a vector variable. State the condition satisfied by f at a point ξ where f is differentiable and has a local maximum or minimum. Show that $f(x_1, x_2) = x_1 x_2$ has a stationary point at $(0,0)$ which is not a local maximum or minimum.	**3**
2	Show that the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1 e^{-x_2}$ is twice differentiable at every point and determine $f''(x_1, x_2)$.	**3**
3	State the mean value theorem for a real-valued function of a vector variable. Show that there is no $\xi \in (0,1)$ such that $f(1) - f(0) = f'(\xi)(1-0)$ in the case of the function $f : \mathbb{R} \to \mathbb{R}^2$ defined by $f(t) = \begin{pmatrix} t^2 \\ t^3 \end{pmatrix}$.	**3**
4	State Taylor's theorem of order 2 for a real-valued function f of a vector variable. What does it reduce to at a stationary point ξ?	**3**
5	Define a positive definite and a negative definite symmetric matrix. Find the eigenvalues of $H = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and state whether H is positive definite.	**3**
6	Define an affine set and a convex set. Show that every affine set is convex.	**3**
7	Find the stationary point of $f(x_1, x_2) = x_1^2 + x_2^2 - 2x_1 + 4x_2$ and classify it using the Hessian matrix.	**3**
8	Define a convex combination of points and the convex hull $\mathbf{conv}\, C$ of a set C. Describe $\mathbf{conv}\{(0,0), (2,0), (0,2)\}$ in $\mathbb{R}^2$.	**3**
9	Show that the intersection of any family of convex sets in $\mathbb{R}^n$ is convex.	**3**
10	Define a cone and a proper cone. For a proper cone K, what does $x \preceq_K y$ mean? Write this out for the nonnegative orthant $K = \mathbb{R}^n_+$.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	A function $f : \mathbb{R}^n \to \mathbb{R}$ is defined by $f(x) = (Ax + a)^T (Bx + b)$, where A and B are $m \times n$ matrices, x is an $n \times 1$ column vector and a, b are $m \times 1$ column vectors. Find a system of linear equations satisfied by the local maxima and minima of f. [Hint: Use the method of the least squares example.]	**6**
12	Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x_1, x_2) = \dfrac{x_1^3 x_2}{x_1^2 + x_2^2}$ for $(x_1, x_2) \neq (0,0)$ and $f(0,0) = 0$. Show that all second order partial derivatives of f exist at $(0,0)$ and that $f_{12}(0,0) \neq f_{21}(0,0)$. Explain why this does not contradict the symmetry of f'' when the second order partial derivatives are continuous.	**6**
13	State and prove the mean value theorem for a real-valued function f of a vector variable, where f is continuous on the line segment L joining a and b and differentiable at each point of L other than a and b.	**6**
14	Suppose that λ is an eigenvalue of an $n \times n$ symmetric matrix M with real entries. Using the results quoted in this chapter, prove that $\lambda^2 \leq \sum_{i=1}^{n} \sum_{j=1}^{n} m_{ij}^2$.	**6**
15	Classify the stationary points of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by $f(x_1, x_2) = x_1^2 x_2 + x_2^2 x_1 - x_1 x_2$ using the second derivative.	**6**
16	Find the stationary points of $f(x_1, x_2, x_3) = x_1 x_2 + x_2 x_3 + x_3 x_1$. Find the eigenvalues of the Hessian matrix and classify each stationary point.	**6**

17	Let $C \subseteq \mathbb{R}^n$ be a convex set, with $x_1, \ldots, x_k \in C$, and let $\theta_1, \ldots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0$, $\theta_1 + \cdots + \theta_k = 1$. Show that $\theta_1 x_1 + \cdots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k.) Hint: Use induction on k.	**6**
18	Let $f(x) = Ax + b$ be an affine function from $\mathbb{R}^n$ to $\mathbb{R}^m$. Show that (i) the image $f(S)$ of a convex set $S \subseteq \mathbb{R}^n$ and (ii) the inverse image $f^{-1}(C)$ of a convex set $C \subseteq \mathbb{R}^m$ are convex.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	(a) Show that the convex hull of a set S is the intersection of all convex sets that contain S. (b) Let $K \subseteq \mathbb{R}^n$ be a proper cone and define $x \preceq_K y$ if and only if $y - x \in K$. Prove that $\preceq_K$ is reflexive, transitive and antisymmetric, and that $x \preceq_K y$ and $u \preceq_K v$ imply $x + u \preceq_K y + v$.	**10**
20	(a) State Taylor's theorem of order 2 for a real-valued function f of a vector variable. If ξ is a stationary point of f, show that $f(\xi + h) - f(\xi) = \frac{1}{2} h^T f''(\eta) h$ for some η on the line segment between ξ and $\xi + h$. (b) Find and classify the stationary points of $f(x_1, x_2) = x_1^3 - 3x_1 x_2^2 + x_2^4$. At the point where the second derivative test fails, decide directly whether f has a local maximum or minimum.	**10**